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1 Fundamentals

1.1 Preamble + notation stuff

A mathematical statement is a sentence like $1 + 1 = 2$ that can be true or false. For example $2 + 2 = 5$ is a statement, just not a true statement.

A proof is a sequence of true statements establishing a conclusion. You've all seen them before.

We prove things to know they are true and why. When we prove things, we need to be careful to prove the right things. We should not prove "if B then A" instead of "if A then B".

A lemma is a statement we prove first then use for a larger proof.

Notation:

The set of natural numbers $\{1, 2, 3, \dots\}$ is denoted by $\mathbb{N}$

The set of integers $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ is denoted by $\mathbb{Z}$

The set of rational numbers (Numbers that can be written as an element of $\mathbb{N}$ divided by a non-zero element of $\mathbb{Z}$) is denoted by $\mathbb{Q}$. However, the ancient Greeks realized that not all numbers were of this form. The length of the diagonal of a unit square is $\sqrt{2}$, which is a real number that when you square it you get 2, and this provably cannot be written as a/b where a and b are integers. Apparently, and this may be a legend, someone in ancient Greece was drowned as a punishment for saying that irrational numbers existed, because all numbers were thought to be rational.

The set of real numbers is denoted by $\mathbb{R}$.

The set of complex numbers is denoted by $\mathbb{C}$.

The square root of 2 is a solution to the equation $x^2 - 2 = 0$. A natural question is whether every real number is the solution to a polynomial with integer coefficients. It turns out that this is not the case – There exist transcendental numbers, which are defined as numbers where this does not hold. We will prove this later.

One principle with proofs is that if we can find a counterexample to a statement, we immediately know that it is false.

Example. We know that the statement “If 9 divides n^2 then 9 divides n ” is false because 9 divides 3^2 but 9 does not divide 3.

More notation:

If A is a mathematical statement, then we write $\neg A$ to mean “Not A”.

If A and B are mathematical statements, we write $A \vee B$ for “A or B” and $A \wedge B$ for “A and B”. We write $A \Rightarrow B$ for “A implies B”, $A \Leftarrow B$ for “B implies A”, and $A \Leftrightarrow B$ for when A and B imply each other, which means A and B are equivalent. This is often written as A iff B or A if and only if B.

If $A \Rightarrow B$, then a counterexample is an example of $A \wedge \neg B$. In plain english, this is saying that a counterexample to A implies B is a situation where A is true and B is not true.

Also, if we want to prove $A \Rightarrow B$ we can often prove instead that $\neg B \Rightarrow \neg A$. This is because if A implies B, then whenever B is false, it must be that A is also false, since A implies B. Similarly, $\neg B \Rightarrow \neg A$ implies $A \Rightarrow B$ so the 2 notions are indeed equivalent. This is called the contrapositive

Example. Here n is an integer. If 2 divides n^2 then n is even.

Proof. We will prove the contrapositive, ie if n is odd then n^2 is odd.

Since n is odd, there is an integer k such that $n = 2k + 1$. Then $n^2 = (2k + 1)^2 = 2(2k^2 + 2k) + 1$ by simple algebra, which is also odd. So done.

□

More notation:

$\forall$ means “for all”

$\exists$ means “there exists”

s.t. means “such that”.

Recall that for sets, $A \cup B$ is the union, ie the set of everything in A or B, $A \cap B$ is the intersection, ie all elements in A and B, and $a \in A$ means a is an element of the set A, and $A \subseteq B$ means A is a subset of B, and $A \subset B$ means A is a proper subset of B (ie A is a subset of B and A does not equal B).

If $a < b$ and a and b are real numbers, The interval (a, b) is all numbers between a and b not including a and b. This is called an open interval. $[a, b)$ is the same interval with a included. $(a, b]$ is the same interval with b included. $[a, b]$ is the same interval with both a and b included, ie a closed interval.

Example:

$\forall x \in \mathbb{R} \exists y \in \mathbb{R} : y < x$ means that for all real numbers x there exists a real number y such that y is less than x.

You have to be careful about the order of these quantifiers. The statement

$\exists y \in \mathbb{R} (\forall x \in \mathbb{R} : y < x)$ is saying that there exists a real number y that is less than all real numbers x, which is obviously

false (if there were such a y , the case when for example $x = y - 1$ causes a contradiction), while the above statement is obviously true (as you can pick for example $y = x - 1$).

Also sets are collections of mathematical objects. They are said to be equal if they have the same elements. This is obvious

[Lecture 1 ends]

When I say a statement, I need to unpack the hidden meaning. For example, if I say that $x^2 - 5x + 6 = 0$ has solutions $x = 2$ and $x = 3$, I am implicitly saying

1. Those are solutions
2. Those are the only solutions

Also, I will now give a false proof that will give some important lessons.

Claim (which is obviously nonsense): Every positive real number is greater than or equal to 1.

“Proof”: Let r be the least positive real. Then either $r < 1$, $r > 1$ or $r = 1$. If $r < 1$, then $0 < r^2 < r$, contradicting the assumption that r is the smallest positive real. Since this is a contradiction, we cannot have $r < 1$. If $r = 1$ then $\frac{1}{2}$ is smaller so that is not the case, and if $r > 1$ then 1 is smaller so r is also not the smallest real number, so we don't have the smallest real number in either of these cases.

The reason this is wrong is because:

1. What I have really proven is that the smallest real number does not exist, so we cannot say “Let r be the least positive real”.
2. For this reason, it is important that we carefully justify everything, including that objects we claim properties of actually exist in the first place.

1.2 Truth tables

I will show this using an example.

Let's make a truth table for statements A and B:

A	B	$A \vee B$	$A \wedge B$	$\neg A$	$A \Rightarrow B$	$B \Rightarrow A$	$A \Leftrightarrow B$
False	False	False	False	True	True	True	True
False	True	True	False	True	True	False	False
True	False	True	False	False	False	True	False
True	True	True	True	False	True	True	True

Now hopefully columns 3-5 are obvious. In column 6, we have that if A is false, then A implies B is true because A implies B is saying that B is true if A is true but that if A is false it does not matter so the statement is vacuously true. Similar for column 7. Then the last column is saying that A implies B and B implies A.

In fact, now one can check using a truth table like this that indeed “A implies B” is the same as “Not B implies Not A”.

Notation: If A and B are sets, $A \setminus B$ means the set of all elements of A not contained in B.

Proposition. The following are true

1. Union is associative, ie $A \cup (B \cup C) = (A \cup B) \cup C$
2. Intersection is associative, ie $A \cap (B \cap C) = (A \cap B) \cap C$
3. Union and intersection are distributive, ie $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Proof. You can easily verify these by shading in the appropriate areas of a venn diagram, or by checking using a truth table. I'll do #3 as an example.

X in A	X in B	X in C	X in $B \cap C$	X in $A \cup (B \cap C)$	X in $A \cup B$	X in $A \cup C$	X in $(A \cup B) \cap (A \cup C)$
No	No	No	No	No	No	No	No
No	No	Yes	No	No	No	Yes	No
No	Yes	No	No	No	Yes	No	No
No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Yes	No	No	No	Yes	Yes	Yes	Yes
Yes	No	Yes	No	Yes	Yes	Yes	Yes
Yes	Yes	No	No	Yes	Yes	Yes	Yes
Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

As you can see, columns 5 and 8 match, as required.

□

[Lecture 2 ends]

We have the following additional 2 identities which can be verified from the table:

1. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
2. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

What this means that when we're in A, (Not in [B or C]) is the same as ([Not in B] and [Not in C]), and that (Not in [B and C]) is the same as (Either [not in B], or [not in C]) which makes sense.

Also, for statements A and B,

$$A \wedge B = \neg(\neg A \vee \neg B), \text{ and}$$

$$A \vee B = \neg(\neg A \wedge \neg B)$$

The reason why is the same idea as the two set identities above. These are variations of De Morgan's laws.

Also, for set identities like $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ we can prove it by showing that $A \cap (B \cup C)$ is a subset of $(A \cap B) \cup (A \cap C)$ and that $(A \cap B) \cup (A \cap C)$ is a subset of $A \cap (B \cup C)$.

Example. If $x \in A \cap (B \cup C)$ then $x \in A$ and $x \in B \cup C$ so either x is in A and B or x is in A and C, which means $x \in (A \cap B) \cup (A \cap C)$.

Conversely, if $x \in (A \cap B) \cup (A \cap C)$ then x is either in A and B or x is in A and C. Either way, x is in both A, and either B or C, so $x \in A \cap (B \cup C)$.

1.3 Set stuff

Notation:

If $A_1, A_2, \dots, A_n$ are sets, or more generally if we have an infinite family of sets, we can write $\bigcap_{r=1}^n A_r$ for the intersection of the sets i.e. the set of elements in all the A's.

Similarly, we can write $\bigcup_{r=1}^n A_r$ for the union of the sets, ie the set of elements in any of the A's.

Given an index set I and a collection of sets denoted by A_i we can write $\bigcup_{i \in I} A_i$, and same for intersections, kind of like we do sometimes for sums.

Definition. Given sets A and B we can form their cartesian product $A \times B$, which is the set of all ordered pairs (a, b)

with a in A and b in B .

We could also have sets of ordered triples like (a, b, c) . In fact, the set of ordered pairs (x, y) with x and y real numbers corresponds exactly to the standard two dimensional plane.

Definition. The power set of a set A is the set of all subsets of A . For example, the power set of the set $1, 2$ is the set $\{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

Note: We can define subsets of other sets by saying $X = \{x \in A \text{ such that something}\}$, but we cannot define $X = \{x \text{ such that something}\}$. There is no universal set of any kind, and we should only construct sets from known sets. The reason is to avoid stuff like defining $Y = \{x : x \text{ is a set and } x \notin x\}$ as this is known as Russell's paradox. If $Y \in Y$ then by definition $Y \notin Y$ but if $Y \notin Y$ then by definition $Y \in Y$.

Also a finite set has size n if it has n elements. A set is defined to be finite if it can be put into one-to-one correspondence with $\{1, 2, 3, \dots, n\}$ for some n .

1.4 Functions

Definition. A function from a set A to a set B is a rule that assigns each element of A to a single element of B , and yeah we've seen before what a function is. We can also think of a function as a subset of $A \times B$ consisting of all the ordered pairs of the form $(a, f(a))$ for a in A . In fact if $A = B = \mathbb{R}$ then this interpretation is exactly the graph of the function.

We can write a function like $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto x^2$. This is a function. However, $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto x^{-1}$ is not a function since 0 is not mapped to anything, and $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto \pm\sqrt{|x|}$ is not a function because each input except for 0 maps to 2 outputs.

Definition. If we have a function that maps elements a in a set A to elements b in a set B , then the image of a subset $C \subseteq A$ is the set $\{f(c) : c \in C\}$. The pre-image of a subset $D \subseteq B$ is the set of elements x in A such that $f(x) \in D$.

Example. If $f(x) = x^2$ then the image of 2 is the set consisting of 4 , but the pre-image of 4 is the set consisting of 2 and -2 .

Also, we can define images and pre-images of sets in the way you would expect. For example, if $f(x) = x^2$ then the image of $(2, 3)$ is $(4, 9)$ and the pre-image of $(4, 9)$ is $(-3, -2) \cup (2, 3)$. Notice that the pre-image of an image is not the original thing.

[Lecture 3 ends]

Example. The image of $(-1, 4)$ under the function x^2 is $[0, 16)$ and the pre-image of $(-1, 4)$ under that same function is $(-2, 2)$.

Definition. An ordered pair is a set (a, b) , or $\{\{a, b\}, \{a\}\}$ with the second element to specify which comes first. We can also have ordered triples, like (a, b, c) , or more.

Definition. A function is injective if every output that is mapped to is only mapped to by a single input. Equivalently, f is injective if $f(b) = f(a)$ implies $b = a$. These functions are nice because we can cancel them while doing algebra without running into issues.

Definition. A function is surjective if every output in the codomain gets mapped to at least once.

Figures 1 and 2 illustrate this.

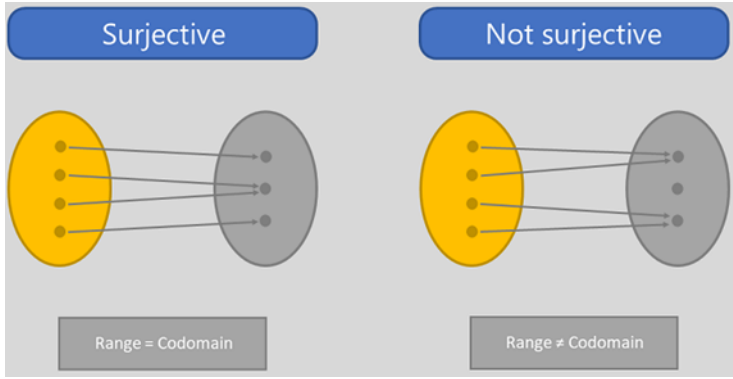


Figure 1

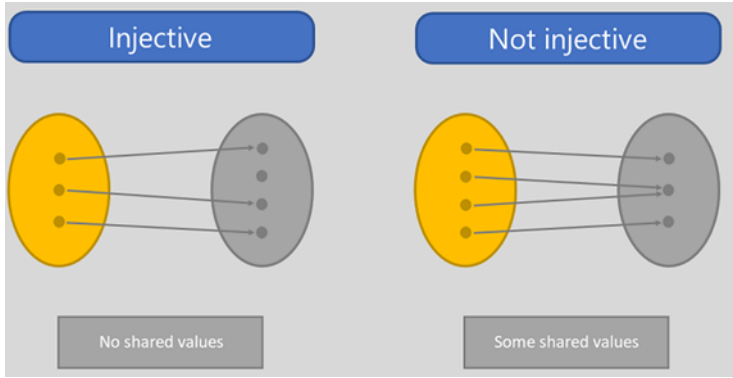


Figure 2

Definition. A bijection is a function which is both injective and surjective. This is a one-to-one mapping where every output is mapped to by exactly one input.

Definition. A permutation is a bijection from a set X to itself. This can be thought of as re-ordering the elements of X .

Note that it is important to specify the domain and codomain of a function. This is often implied, but a question like “is x^2 injective” technically speaking is meaningless, since the answer is yes in the natural numbers and no in the real numbers.

We denote the size of a set A using $|A|$ or $\#A$. We observe some facts which are obvious for finite sets:

1. $|A| < |B|$ if there is no surjection from A to B
2. $|A| > |B|$ if there is no injection from A to B
3. By definition, $|A| = |B|$ if and only if there is a bijection from A to B , and if furthermore we assume the sets are both finite then any function from A to B is injective if and only if it is surjective.

Remark. A being a subset of B that is not B itself only implies A is smaller than B if A and B are finite. For example: if A is the even integers and B is the integers, then A is a subset of B , but they are the same size as simply mapping each element of A to the element in B equal to half of that element gives a bijection.

In level 6, we proved that $|\mathbb{R}| > |\mathbb{N}|$ using Cantor’s diagonal argument.

Definition. A function X to X where every element is mapped to itself is called the identity function.

A sequence can be interpreted as a function $a_n : \mathbb{N} \rightarrow \mathbb{R}$.

Addition $(+)$ can be thought of as a function from $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ since it takes in two natural numbers and outputs a third. In general, a binary operation is a function like this on a set X from $X \times X \rightarrow X$.

Definition: For a set X and a subset Y of X , the indicator function, denoted $i_Y(A)$ is defined to be 1 if A is in Y and 0 otherwise.

Here are some facts, each of which are easy to see using a truth table:

1. If the functions i_A and i_B are the same then A and B are equal, and the converse of this is also true.
2. $i_{A \cap B} = i_A i_B$. This can be verified by going through the different possibilities.
3. $i_{A \cup B} = i_A + i_B - i_A i_B$. Again one can easily verify this.
4. $i_{X \setminus A} = 1 - i_A$ (for everything in X in the first place)

Another way to see (iii) is the following:

$$i_{A \cup B} = i_{X \setminus ((X \setminus A) \cap (X \setminus B))} = 1 - i_{(X \setminus A) \cap (X \setminus B)} = 1 - i_{X \setminus A} i_{X \setminus B} = 1 - (1 - i_A)(1 - i_B) = i_A + i_B - i_A i_B$$

[Lecture 4 ends]

We can have composite functions. This is denoted by $f \circ g$ to mean $f(g(\text{input}))$. This is clearly associative: $(f \circ g) \circ h = f \circ (g \circ h)$

Definition. A function f is invertible if there exists a function g such that $f \circ g = g \circ f = \text{identity}$, and it is the case that if f goes from A to B then g goes from B to A.

Example. the function $x + 1$ on the natural numbers is not invertible because if we try to make $x - 1$ the inverse, $g(1)$ would have to be 0, as otherwise you couldn't add 1 and get 1, so $f \circ g$ could not be the identity.

If $g \circ f = \text{identity}$ then g is called a left inverse of f . If this is the case, f must be injective, since if $f(a) = f(b)$ then $g(f(a)) = g(f(b))$ so $a = b$ since $g \circ f = \text{identity}$, so we satisfy the injectivity criterion of being able to cancel the f .

Conversely, if f is injective and its domain is non-empty then there is a left inverse. We construct this as follows: For each element in the range of f , send it back. For elements not in the range, pick any element to send it to (since f is non-empty). One can easily check that this works: $g(f(a))$ has a sent to something in f 's range then sent back.

If we have a right inverse, i.e. $f \circ g = \text{identity}$ then for every $a \in \text{Domain}(g)$ we have $a = f(g(a))$ so a lies in $\text{Range}(f)$. Since we are assuming that $\text{Range}(f) = \text{Domain}(g)$ it follows that f is surjective.

Conversely, if f is surjective, define g to take every element in the range of f back to any element (which we pick)¹ that maps to that element in the range. Doing so will give a right inverse.

Therefore, a function is invertible exactly when it is bijective.

$f^{-1}(A)$ where A is a set is often used as notation for the pre-image of A.

1.5 Relations

Definition. A relation on a set X can be thought of as a subset R of $X \times X$. We write aRb if (a, b) is in R .

Examples of relations on the natural numbers:

1. aRb if a and b have the same last digit
2. aRb if $a < b$
3. aRb if a does not equal b
4. aRb if $a = b = 1$
5. aRb if $|a - b| < 3$

¹Yes I know technically we need the axiom of choice, good thing the axiom of choice is true.

Definition. A relation is reflexive if every element is related to itself (ie aRa for all a).

Definition. A relation is symmetric if for every a and b in our set aRb implies bRa .

Definition. A relation is transitive if for every a, b, c in our set aRb and bRc implies aRc .

Definition. A relation is an equivalence relation if it satisfies all three of the properties above.

Here is a table based on the 5 examples above which you should stare at until you can convince yourself that it is correct:

	Example 1	Example 2	Example 3	Example 4	Example 5
Symmetric	Yes	No	Yes	Yes	Yes
Reflexive	Yes	No	No	No	Yes
Transitive	Yes	Yes	No	Yes	No

[Lecture 5 ends]

Theorem. Theorem: If $\sim$ is an equivalence relation on a set X we have a bunch of pairwise disjoint subsets of X whose union is all of X called equivalence classes such that two elements are related if and only if they are in the same class.

Proof. Suppose we have such sets. One easily checks that a relation defined as being satisfied if two elements are in the same class is an equivalence relation. Now suppose we have an equivalence relation. Suppose $[x]$ is the set of elements y that are related to x , ie $y \sim x$ or $x \sim y$ which are the same thing by the symmetry property. And $x \sim x$ by reflexivity, so every element is in its own class, and we just have to prove that the classes are pairwise disjoint, in the sense that for each z , either $[x] = [z]$ or $[x] \cap [z]$ is the empty set. Suppose there is an element y in both $[x]$ and $[z]$, then y is related to both x and z , which by transitivity means x must be related to z . If an element w is in $[z]$ then w is related to z and thus related to x so it is also in $[x]$. Therefore, $[z]$ is contained in $[x]$ and by the same logic $[x]$ is contained in $[z]$, so if they have non-empty intersection they are the same.

□

We define the quotient of X by an equivalence relation R as $X/R =$ The set of equivalence classes.

In example 1 from the previous lecture, the equivalence classes were the set of natural numbers with a particular last digit.

The map (which just means function) $X \rightarrow X/R$ that sends an element of X to its own equivalence class called the projection map.

Example. On the set $(\mathbb{Z} \setminus 0) \times \mathbb{N}$ we can define an equivalence relation by $(a, b)R(c, d)$ if $ad = bc$, or equivalently $\frac{a}{b} = \frac{c}{d}$, which is an equivalence relation since it can be easily seen that the equivalence classes correspond exactly to the rational numbers.

More on binary operations:

A binary operation $.$ is commutative if it is always true that $A.B = B.A$

It is associative if it is always true that $A.(B.C) = (A.B).C$

A binary operation $.$ is distributive over a binary operation $\sim$ if it is always true that $A.(B \sim C) = (A.B) \sim (A.C)$ and $(A \sim B).C = (A.C) \sim (B.C)$, i.e. they interact the same way multiplication interacts with addition.

2 The natural numbers

2.1 Peano arithmetic

We define the set of natural numbers $\mathbb{N}$ to have an element which we call 0. It has a function called S that we think of as adding 1 such that S is injective and every element that is not 0 is S (something in $\mathbb{N}$), and 0 is not S (anything). Also, so

that we ensure we only have the natural numbers (as something like the set $\mathbb{N} \cup (\mathbb{Z} + 0.5)$ satisfies these axioms), we have induction as an axiom: If A is a set with $0 \in A$ and the property that if $a \in A$ then $s(a) \in A$ then $A = \mathbb{N}$.

We define addition as $n + 1 = S(n)$ [A1] and $n + S(m) = S(n + m)$ [A2] and multiplication defined as $n * 1 = n$ and $n * S(m) = n * m + n$. We define 0 such that $a + 0 = a$ always. In theory, we can prove all the obvious properties of these operations. Although we do not need to do that in this course as it is ok to do math considering associativity of addition and multiplication as well as the distributive property to all be given,

Example. I will prove that $(a + b) + c = a + (b + c)$ for every a, b, c from the Peano axioms to demonstrate how these proofs generally go.

If $c = 0$,

$$(a + b) + 0 = a + b = a + (b + 0)$$

Each equation follows from [A1]

Now for the induction hypothesis we will assume that for some c , we indeed have

$$(a + b) + c = a + (b + c)$$

We want to prove that this also holds for $S(c)$. By [A2] and the induction hypothesis, we get

$$\begin{aligned} (a + b) + S(c) &= S((a + b) + c) \\ &= S(a + (b + c)) \\ &= a + S(b + c) \\ &= a + (b + S(c)) \end{aligned}$$

This completes the induction step.

[Lecture 6 ends]

2.2 Well ordering and induction

Definition. An ordering on a set is a way to compare elements, where some pairs satisfy exactly one of $a = b$, $a < b$ and $a > b$, and just to make sure it is actually an ordering in the way we expect, we have that ordering is transitive ($a < b$ and $b < c$ implies $a < c$, and similarly for $>$).

Definition. An ordering is said to be a Total Ordering if any two elements can be compared. An example of a partial ordering is an ordering on sets such that $A < B$ if A is a subset of B and not equal to B , which satisfies the rules above but 1 and 2 cannot be compared. Otherwise it is said to be a partial ordering.

The axiom of induction given above is called the weak principle of induction (*WPI*). The strong principle of induction (*SPI*), which we saw in the level 4 proof for uniqueness of prime factorization and polygon angle sums, states that if it is the case that $\{0, 1, 2, 3, \dots, n\}$ being a subset of a set A implies that $n + 1$ is in the set A , and 0 is in the set A then all natural numbers are in the set A . We will prove some things that seem useless but are useful because they can apply to other sets with other axioms defined.

This is confusing, as we have a (Statement 1 implies Statement 2) implies (Statement 3) situation.

Theorem. WPI implies SPI

Proof. Apply WPI to the set of numbers n such that $\{0, 1, 2, 3, \dots, n\}$ is a subset of A .

□

Theorem. SPI implies WPI

Proof. If n is in A implies $n+1$ is in A then clearly $\{1, 2, 3, \dots, n\}$ being a subset of A also implies $n+1$ is in A so by strong induction A has all the natural numbers.

□

Definition. A total ordering on a set is called well-ordered if it has the property that every non-empty subset has a least element with respect to this ordering.

The well-ordering principle (WOP) states that the usual ordering on $\mathbb{N}$ is a well-ordered. This is obvious, and we will prove it shortly. An example of a total ordering that is not a well ordering is $\mathbb{Z}$ with the usual ordering: The set of, say, even integers does not have a least element: they keep going down forever. However, we can define an ordering on $\mathbb{Z}$ that is a well ordering, where we say the order of the elements is $(0, -1, 1, -2, 2, \dots)$

Theorem. SPI implies WOP

Proof. Suppose there is no least element of a subset P of a set where SPI holds. Then consider the set Q of elements that are not in P : 1 is in Q otherwise 1 would be the minimal element of P . 2 is also in Q or else 2 would be a minimal element of P , and so on. In general, $\{1, 2, 3, \dots, n\}$ being in Q implies $n + 1$ is in Q otherwise that would be the least element of P . So by strong induction this implies Q is the whole set.

□

Example. We can prove using the well ordering principle that every number that is ≥ 2 can be factored into primes (although as we know proving uniqueness is harder). Let S be the set of counterexamples, then S has a least element m . If m is prime, this is a contradiction, and if not then by definition we can write m as ab , where a and b are less than m and thus can be factored into primes, so we also have a contradiction.

2.3 The binomial theorem

Theorem. A power set of a set with size n has 2^n elements.

Proof. For each of the n elements, we can either put it or not put it in a subset which there are 2 ways to do. Choosing between 2 things to do n times gives a total of 2^n choices.

□

Corollary. $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$ because we can interpret LHS as the number of subsets with 0 elements + the number with 1 elements + the number with 2 and so on.

As another corollary, we get $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ since we can either choose or not choose the last element, and then in each case we have $k-1$ or k remaining choices from the first $n-1$ elements.

We can obtain Pascal's triangle by starting with 1's on the sides and having each element being the sum of the two above it. See figure 3.

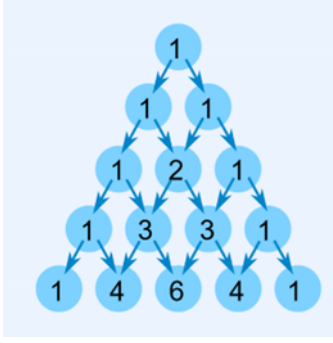


Figure 3

Because of this, the number in each circle counts the number of ways to get there, since we can get there from either of the circles above it.

In fact, the k 'th number in the n 'th row is $\binom{n}{k}$. This is because we have to take k right arrows in order to get there and there are n rows that can happen in and we need to choose k of those rows to take our right turn, if that makes sense.

The formula $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ as well as the binomial theorem is known from A level. When n is large and k is fixed, this is well approximated by $\frac{n^k}{k!}$, since it can be written as $\frac{n(n-1)(n-2)\dots(n-k+1)}{k!}$. We will give another proof of the binomial theorem soon.

2.4 The inclusion-exclusion principle

Observe that $|A \cup B| = |A| + |B| - |A \cap B|$

Let's try to count $|A \cup B \cup C|$. If we write this as $|A| + |B| + |C|$ we double count the stuff in the intersections.

If we then write $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A|$

This is closer but now we would not be counting anything in $A \cap B \cap C$. The correct expression is

$$|A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$$

The general theorem is as follows:

Theorem. Let $S_1, S_2, S_3, \dots, S_n$ be finite sets, then

$$|S_1 \cup S_2 \cup S_3 \cup \dots \cup S_n| = \sum |S_i| - \sum |S_{i_1} \cap S_{i_2}| + \sum |S_{i_1} \cap S_{i_2} \cap S_{i_3}| - \sum |S_{i_1} \cap S_{i_2} \cap S_{i_3} \cap S_{i_4}| + \dots$$

Where the j 'th sum ranges over all $1 \leq i_1 < i_2 < \dots < i_j \leq n$.

We can alternatively write the theorem as the following functions being equal at every input value

$$\sum i_{S_1 \cup S_2 \cup S_3 \cup \dots \cup S_n} = \sum i_{S_A} - \sum i_{S_A} i_{S_B} + \sum i_{S_A} i_{S_B} i_{S_C} - \dots$$

Proof. Let $x \in (S_1 \cup S_2 \cup S_3 \cup \dots \cup S_n)$. Suppose that x lives in exactly k of the sets. Then the number of distinct pairs of sets that x lives in is $\binom{k}{2}$, and the number of distinct triples that x lives in is $\binom{k}{3}$ and so on. Therefore, the value of $\sum i_{S_A} - \sum i_{S_A} i_{S_B} + \sum i_{S_A} i_{S_B} i_{S_C} - \dots$ is exactly equal to $k - \binom{k}{2} + \binom{k}{3} - \binom{k}{4} + \dots - (-1)^k \binom{k}{k} = 1 - \left(1 - k + \binom{k}{2} - \binom{k}{3} + \binom{k}{4} - \dots + (-1)^k \binom{k}{k}\right)$ By the binomial theorem, this is the same as $1 - (1 + (-1))^k$ which is 1 as we want.

Note that if $x \notin (S_1 \cup S_2 \cup S_3 \cup \dots \cup S_n)$ then all terms in the sum go to 0 so the equality becomes trivial.

□

2.5 Properties of prime numbers and factorization

Notation: We say $a|b$ to mean a divides b , meaning there is an integer c such that $ac = b$. We can also say that a is a divisor or factor of b , or b is a multiple of a .

Definition. A composite number is a positive integer that is not prime, except for 1 which is neither prime nor composite.

We know that induction can be used to prove statements like the statement that every number is the product of primes in a unique way up to ordering. Recall that the proof that there are infinitely many primes is a classic argument (If there are finitely many, multiply them together and add 1 to get a number with a factor not already in the list).

Proposition. Let n and k be positive integers, then we can divide n by k and get a remainder less than k , meaning we can write $n = qk + r$ where r is non-negative less than k . This is unique as if we change q r is no longer in that range.

Hopefully you can see that this is clearly true, if not just divide both sides by k , and then use the fact that any real number has an integer part and a fractional part less than 1.

Consider two numbers a and b , then they have a highest common factor (hcf), also called a greatest common divisor (gcd).

Now we will demonstrate Euclid's algorithm by example to show how to find the highest common factor of two numbers.

Example. Suppose we want to find $\text{hcf}(420, 792)$, which we will call c .

This procedure will resemble what we did for polynomials in Level 4 to prove partial fraction decomposition.

Clearly, c divides 792 and 420, so it also divides their difference which is 372. Our new number (372) is constructed as the remainder of the bigger number (a or 792) when divided by the smaller number (b or 420). This is equal to $r = a - qb$ by rearranging the equation above, therefore any common factor of a and b also divides r and the converse is true as well. Specifically, c is $\text{hcf}(420, 372)$ because if there was a larger common factor of 420 and 372 it would be a common factor of 420 and 792 as well which contradicts c being the highest. At each step when we repeat this, the remainder will be smaller than our smaller number so we will eventually reach 0 in finitely many steps since everything is an integer.

Now we do the same for 372 and 420: We find the difference which is 48, then $\text{hcf}(372, 48) = c$. Now we keep subtracting 48 from 372 until we can't anymore, then the final remainder is 36, so we get $\text{hcf}(36, 48) = c$. Now continuing, we get $\text{hcf}(36, 12) = c$ then $\text{hcf}(0, 12) = c$, so $c = 12$.

Corollary. Any common factor of a and b divides $c := \text{gcd}(a, b)$

Proof. By doing Euclid's algorithm, we can see that all common factors of a and b are common factors of 0 and c since we explained above that the common factors stay the same at each step, thus any common factor of a and b is a common factor of 0 and c , and therefore divides c .

Therefore, we can define $\text{gcd}(a, b)$ as the unique positive integer dividing a and b such that any common factor of a and b divides $\text{gcd}(a, b)$, as it is now clear that this exists and is unique.

□

Figure 4 shows a nice visual representation of Euclid's algorithm.

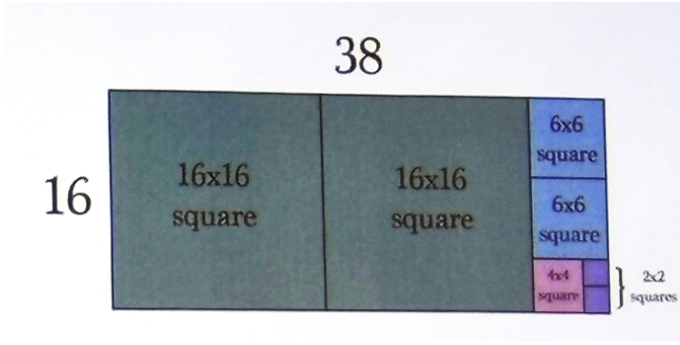


Figure 4

Definition. We say numbers are coprime or relatively prime if their highest common factor is 1.

Geometric interpretation: if we have an $A * B$ grid rectangle with A and B coprime its diagonal will not intersect any grid corners.

Now we will do something similar to what we did for polynomials in Level 4:

Example. An example of the euclid algorithm on coprime numbers (52 and 87) is as follows:

$$87 = 1 * 52 + 35$$

$$52 = 1 * 35 + 17$$

$$35 = 2 * 17 + 1$$

Now we get to 1. Let's work backwards from 1:

$$1 = 35 - 2 * 17$$

$$1 = (52 - 17) - 2 * (52 - 35)$$

$$1 = (52 - (52 - 35)) - 2 * (52 - 35)$$

$$1 = 52 - (52 - (87 - 52)) - 2 * (52 - (87 - 52))$$

Collecting like terms will give

$$1 = 3 * 87 - 5 * 52$$

In general, we can always do a procedure like this: If A and B are coprime we can run Euclid's algorithm and work backwards from 1 to get integers x and y with $Ax + By = 1$. More generally, we can find integers x and y with $Ax + By = \gcd(A, B)$. The fact that we can do this is called Bezout's identity. This is exactly what we did for polynomials in Level 4 before hinting at this when we showed that a linear combination of two polynomials can equal their highest common factor. This proves existence of such x and y and also gives a nice way to find them.

Alternative proof of Bezout's identity:

Let h be the least positive integer that can be written as $Ax + By$ for integers x and y . h exists by the well ordering principle and the fact that the set is non-empty since it contains $A + B$ for example. Clearly, all common factors of A and B are a factor of $Ax + By$ and thus h . Now suppose that h does not divide A : Then we can write $a = qh + r$ with $0 < r < h$ and q an integer. So

$$r = A - qh = A - q(Ax + By) = (1 - qx)A + (-qy)B$$

contradicting the definition of h since h was the smallest positive integer that can be written as $Ax + By$ for integers x and y but r is smaller. Therefore h divides A , and by similar logic h also divides B . Therefore, h is a common factor of A and B and all common factors of A and B have to divide h since $h = Ax + By$, so h is the highest common factor of A and B , so done. This proof tells us that the integers x and y exist but does not give us a way to find them.

Example. Does there exist x and y with $160x + 72y = 33$. I mean, this one is obviously no because the left hand side is even and the right hand side is odd, but the point is such integers exist exactly when the number on the right is a multiple of the highest common factor of 160 and 72, or in general the numbers A and B we start with. The reason is if the right is divisible by $h = \text{hcf}(A, B)$ we can write it as Ch so we can find a solution for when it equals h and multiply it by C . Conversely, if h does not divide the right hand side, then we can write it as $Ch + r$ with $r < h$, but then $Ch + r$ and Ch can both be written as $Ax + By$, so the difference gives r as a solution which is a contradiction.

Proposition. If p is a prime and p divides ab then p divides a or p divides b .

Proof. Suppose p does not divide a . Then $\text{hcf}(p, a)$ is 1 since the only factors of p are 1 and p and p does not divide a . Then there exists integers x and y with $px + ay = 1$, so $pbx + aby = b$. But p divides ab , so therefore p divides the left hand side, and therefore p divides b . □

Corollary. If $p|a_1a_2a_3 \dots a_n$ and p is prime then by a simple induction, $p|a_i$ for some i from 1 to n .

Corollary. Now we can prove uniqueness of prime factorization. If you saw the proof in level 4 on this website, this one is different. Suppose $n = p_1p_2 \dots p_k = q_1q_2 \dots q_l$, for primes p and q , and suppose these primes are in order from least to greatest. Then p_1 divides $q_1q_2 \dots q_l$ and thus divides one of the q 's by the previous corollary. Thus, one of the q 's equals p_1 since p_1 divides it and they are prime. So we can cancel that q and p_1 from both sides, and we can keep doing this and eventually we will be left with $1 = 1$, and so it will become clear that the p 's and q 's must have been the same at the beginning. If we do this by strong induction, then by cancelling one of the p 's we end up using the strong induction hypothesis to deduce uniqueness of the rest.

Note: There exist number systems in which factorization is not unique. An example is the set of numbers $x + i\sqrt{3}y$ with x and y integers. In this system $4 = 2 * 2 = (1 + i\sqrt{3})(1 - i\sqrt{3})$. Therefore the theorem above should not be considered "obvious", since 2 and $1 \pm i\sqrt{3}$ are "prime"² in this world since they cannot be further factored into integers that are not just 1 or -1.

Example. We can easily list all the factors of 2^33^711 : They are just all the numbers

$$\{2^x3^y11^z : 0 \leq x \leq 3, 0 \leq y \leq 7, 0 \leq z \leq 1\}$$

If there were other factors, this would imply a factorisation of 2^33^711 that is different, contradicting uniqueness. In fact, the number of factors of 2^33^711 must therefore be $(3 + 1)(7 + 1)(1 + 1) = 64$, since we multiply together the number of choices for each of the powers. This is how we can find the number of factors of any number. The general formula is the product of one plus each of the prime powers.

Recall that the common factors and common multiples of numbers a and b can be found by taking the prime factorisation of a and b and taking the minimum and maximum of the powers respectively.

Corollary. $\text{Lcm}(a, b)\text{Hcf}(a, b) = ab$ because the power of some p in $\text{Lcm}(a, b)\text{Hcf}(a, b)$ is the larger of the power of p in a and b plus the smaller of the power of p in a and b , which is exactly the power of p in a plus the power of p in b in some unknown order, which is the power of p in ab . Note that every other common multiple of a and b must be a multiple of $\text{Lcm}(a, b)$, since for each p , the exponent of p in the prime factorisation must be at least that of the power of p in $\text{Lcm}(a, b)$. Alternatively, say for example 10 was the lcm of 2 numbers and so was 45 which is not a multiple of 10, then so would $45 - 4 * 10$ contradicting the assumption that 10 is the lcm, essentially the remainder would be a common multiple.

Example. (Another proof of infinite primes by Paul Erdos, possibly my favourite mathematician) Suppose there are k primes $p_1, p_2, \dots, p_k$. Now consider

$$N := p_1^{j_1} p_2^{j_2} p_3^{j_3} \dots p_k^{j_k} = m^2 p_1^{i_1} p_2^{i_2} p_3^{i_3} \dots p_k^{i_k}$$

where each of the i 's are 0 or 1. Now we know that $m \leq \sqrt{N}$, so there are at most $\sqrt{N}$ possibilities for m , so there are at most $\sqrt{N}2^k$ numbers of the form $m^2 p_1^{i_1} p_2^{i_2} p_3^{i_3} \dots p_k^{i_k}$ that are less than N , so for $N > 4^k$ we contradict the fact that every number less than N can be written in this way, so if $N = 4^{k+1}$ there is some number less than N that cannot be written in this way, then this number must have a prime factor not amongst our finite set of primes.

In particular the prime factor we got in the final contradiction is less than 4^{k+1} . This means that the proof above is stronger than Euclid's proof as it tells us that the k 'th prime must come before 4^k .

²I know that in this context, 'prime' has another meaning, but I'm making a point, see the Groups, Rings and Modules sublevel for more details.

3 Modular arithmetic

3.1 Introduction

Definition. We define $a \equiv b \pmod n$ if a and b differ by a multiple of n . If $0 \leq b < n$, b is the remainder when a is divided by n .

Proposition. Let $c \equiv a \pmod n$ and $d \equiv b \pmod n$, then $c + d \equiv a + b \pmod n$ i.e. addition works mod n

Proof. Any number which is the sum of an $a \pmod n$ and $b \pmod n$ number is $(a + xn) + (b + yn) = (a + b) + (x + y)n$ and therefore is an $a + b \pmod n$ number. □

Proposition. Let $c \equiv a \pmod n$ and $d \equiv b \pmod n$, then $c * d \equiv a * b \pmod n$ i.e. multiplication works mod n

Proof. Similar to above: $(a + xn)(b + yn) = ab + (bx + ay + xyn)n$. □

Example. $2a^2 + 3b^3 = 1$ does not have a solution with a, b in $\mathbb{Z}$. The reason is as follows:

Consider $2a^2 + 3b^3 \pmod 3$. Then $3b^3 = 0 \pmod 3$ so we just have to consider $2a^2 \pmod 3$. If a is a multiple of 3, this is 0 and not 1 which is not possible. If a is $1 \pmod 3$, we have $2(3k+1)^2 \pmod 3 = 18k^2 + 12k + 2 \pmod 3 = 3(6k^2 + 4k) + 2 \pmod 3 = 2 \pmod 3$ which is again not $1 \pmod 3$. If a is $2 \pmod 3$, a similar argument shows that $2a^2 + 3b^3 \pmod 3 = 2 \pmod 3$ and therefore $2a^2 + 3b^3$ can never be $1 \pmod 3$ which is a contradiction so $2a^2 + 3b^3 = 1$ must have no integer solutions. Note that if two numbers are the same mod something, we say they are congruent mod that something.

3.2 Solving equations

Example. Lets solve $7x = 2 \pmod{10}$. We can multiply both sides by 3 to get $21x = 6 \pmod{10}$. But then $20x$ is $0 \pmod{10}$ so we can subtract it to get $x = 6 \pmod{10}$.

Definition. Given integers a and b , we say that b is an inverse of $a \pmod n$ if $ab = 1 \pmod n$. We say a is invertible mod n , or a is a unit mod n , if it has an inverse.

From the previous example, 3 and 7 are inverses of each other mod 10. These either come in pairs or a number is its own inverse.

Example. 4 does not have an inverse mod 10, such an inverse would have to be a number that can be multiplied by an even number to get an odd number.

Proposition. If a is a unit mod n then its inverse is unique mod n .

Proof. Suppose there exists 2 inverses b and b' such that $ab = ab' = 1 \pmod n$. We know that $b = bab = bab' = b' \pmod n$, since $ba = 1 \pmod n$. □

Corollary. If a is a unit mod n and $ab = ac \pmod n$ then $b = c \pmod n$. ie we can cancel units. The proof is immediate, just multiply both sides by the inverse of a .

Remark. This is not true for non-units: $4 * 3$ and $4 * 8$ are the same mod 10 but we cannot divide out the 4 because 3 is not congruent to 8 mod 10.

Proposition. Let p be prime. Then every non-zero residue mod p is a unit mod p .

Proof. If a is a non-zero residue mod p then it is coprime to p , so Bezout's identity asserts the existence of integers x and y with $ax + yp = 1$, so x is an inverse to a mod p .

□

In fact, by the same logic, we deduce that a is a unit mod n if and only if a is coprime to n .

Corollary. If a is coprime to n , then the congruence $ax = b \pmod{n}$ has a unique solution $x \pmod{n}$, where $x = b * (\text{Inverse of } a \pmod{n})$.

Example. Because 365 and 7 are coprime, it means that Christmas can fall on any day of the week (if we pretend like leap year technicalities aren't an issue). This is because the equation $365x = k \pmod{7}$ has a solution for any k .

Proposition. If a and n are not coprime, then $ax = b \pmod{n}$ has a solution if and only if b is a multiple of $\text{hcf}(a, n)$

Proof. If there is a solution then for some x $\text{hcf}(a, n)$ must divide $ax - b$ as n divides $ax - b$. But $\text{hcf}(a, n)$ divides a and $ax - b$ and thus must divide b .

Conversely, if b is a multiple of $\text{hcf}(a, n)$ then for the same reason as above Bezout's identity asserts the existence of a solution. If $\text{hcf}(a, n) = d$, then the solution to $ax = b \pmod{n}$ is exactly the solution corresponding to the solution to $\frac{a}{d}x = \frac{b}{d} \pmod{\frac{n}{d}}$ (we can divide like this because the results are all integers) which we know exists.

□

Example. Lets solve $7x = 4 \pmod{30}$. $\text{hcf}(7, 30) = 1$ so 7 has an inverse. It turns out that this inverse is 13 (To calculate it, you could use Euclid's algorithm, or just use a computer - that is likely how the computer would do it), so we get that

$$x = 4 * 13 \pmod{30} = 22 \pmod{30}$$

Furthermore, this solution is unique. Suppose x' is also a solution with $x' \not\equiv 22 \pmod{30}$, then we have that $7x' = 4 \pmod{30}$. But then $7x$ and $7x'$ are the same mod 30, so multiplying by 13 on both sides it follows that x and x' are equal mod 30.

Example. Lets solve $10x = 12 \pmod{34}$, then $\text{hcf}(10, 34) = 2$ so the solutions are the same as the solutions to $5x = 6 \pmod{17}$, multiplying by 7 on both sides gives $x = 8 \pmod{17}$, so this is our solution.

Example. Lets solve the simultaneous equations $x \equiv 1 \pmod{4}$ and $x \equiv 2 \pmod{3}$.

We know that $x = 1, 5$ or $9 \pmod{12}$ and that $x = 2, 5, 8$ or $11 \pmod{12}$ from each of the above equations. Therefore $x = 5 \pmod{12}$ is the only possibility.

However, the simultaneous congruences $x \equiv 1 \pmod{4}$ and $x \equiv 2 \pmod{6}$ do not have a solution because x has to be both odd and even. The first theorem of the next section will tell us a general situation when we can find a solution.

3.3 Several important theorems

Theorem. (Chinese Remainder Theorem) Let m, n be coprime and a, b be integers. Then the simultaneous congruences $x \equiv a \pmod{m}$ and $x \equiv b \pmod{n}$ have a unique solution mod mn .

Proof. By Bezout's identity, there exists integers s and t satisfying $sm + tn = 1$. Therefore $sm = 1 \pmod{n}$, and $tn = 1 \pmod{m}$. Now consider $x = a(tn) + b(sm)$. Then this is congruent to $a \pmod{m}$ and $b \pmod{n}$. To prove uniqueness mod mn , suppose y is also a solution to the simultaneous congruences. Then $y - x$ must be $0 \pmod{m}$ and $0 \pmod{n}$, and since m and n are coprime, $y - x$ is congruent to $0 \pmod{mn}$, and therefore $x = y \pmod{mn}$.

□

This can be extended by induction: If $m_1, m_2, \dots, m_k$ are pairwise coprime (i.e. any 2 from that list are coprime), then the simultaneous congruences

$$x = a_1 \bmod m_1$$

$$x = a_2 \bmod m_2$$

$$\dots$$

$$x = a_k \bmod m_k$$

has a unique solution mod $m_1 m_2 \dots m_k$.

Definition. $\phi(n)$ is the number of integers from 1 to $n - 1$ coprime to n . Equivalently, this function is counting the number of units mod n . We define $\phi(1) = 1$.

Example. $\phi(9) = 6$ since the units are 1,2,4,5,7,8.

When p is a prime, $\phi(p) = p - 1$ since everything from 1 to $p - 1$ is a unit and $\phi(p^2) = p^2 - p$, since there are p non-units.

We will now look at powers mod n .

Example. $2^1 \bmod 7 = 2, 2^2 \bmod 7 = 4, 2^3 \bmod 7 = 1, 2^4 \bmod 7 = 2$ and we're back where we started.

In general, you can see that if we take powers mod a number, we must eventually get back to somewhere we have been to before, since there are only so many results we can land on, so it will eventually repeat since each result only depends on the previous one in the same way. This leads us to the second important theorem of this section.

Theorem. (Fermat's little theorem) Let p be a prime and a not divisible by p . Then $a^{p-1} = 1 \bmod p$.

Proof. Since a is a unit mod p , $ax = ay \bmod p$ if and only if $x = y \bmod p$.

Now consider the numbers $a, 2a, 3a, \dots, (p-1)a$. These must be pairwise incongruent mod p : if $ia = ja \bmod p$, then $(i-j)a$ is divisible by p , but that is not possible because $0 < |i-j| < p$ and a is not divisible by p , and we know that if p divides $(i-j)a$ then it must divide $i-j$ or a which is impossible.. Therefore, $a, 2a, 3a, \dots, (p-1)a \bmod p$ must be $1, 2, 3, \dots, p-1$ in some order.

Therefore, $(a * 2a * 3a * \dots * (p-1)a) = (1 * 2 * 3 * \dots * p-1) \bmod p$. Therefore, $a^{p-1}(p-1)! = (p-1)! \bmod p$. Since $(p-1)!$ is not divisible by p , it is a unit and we can multiply through by the inverse of $(p-1)! \bmod p$ to get the desired result.

□

Proposition. If p and q are primes and not equal to each other, $\phi(pq) = (p-1)(q-1)$

Proof. mod pq , the ones that are not coprime to pq are the p multiples of q and the q multiples of p , but we have double counted the multiple of pq so we need to subtract 1. Then we get the number that are coprime is pq minus the number that are NOT coprime which therefore is $pq - p - q + 1$ which is indeed $(p-1)(q-1)$.

□

The third important theorem of the section is a generalization of Fermat's little theorem (in fact it is Fermat's little theorem for the case when m is prime).

Theorem. (Fermat-Euler theorem) $a^{\phi(m)} = 1 \bmod m$ for a coprime to m .

Proof. The same way as in the previous theorem: Let $n_1, n_2, \dots, n_{\phi(m)}$ be the units mod m . Then since a is coprime to m , if we multiply everything by a we still have units mod m . We in fact have distinct units since if $an_j - an_i = 0 \bmod m$ then $n_j = n_i \bmod m$ so j must equal i , so multiplying by a just changes the order.

Therefore, since all we did was change the order by multiplying everything by a , we have that $a^{\phi(m)}(n_1 n_2 n_3 \dots n_{\phi(m)}) = n_1 n_2 n_3 \dots n_{\phi(m)} \pmod{m}$. Since $n_1 n_2 n_3 \dots n_{\phi(m)}$ is a product of units, it is itself a unit so we can multiply out by the inverse to get $a^{\phi(m)} = 1 \pmod{m}$ as required.

□

To get to another important theorem of this section, we will consider what the value of $(p-1)!$ is mod p if p is a prime, with the exception of $p = 2$. We will do a few examples in order to guess what the result should be.

If $p = 3$, we want $2 \pmod{3}$ which is 2

If $p = 5$, we want $24 \pmod{5}$ which is 4

If $p = 7$, we want $720 \pmod{7}$ which is 6

If $p = 11$, we want $3628800 \pmod{11}$ which is 10

Interestingly, it seems to always be $-1 \pmod{p}$. We will now prove that this is the case.

Lemma. If p is prime (Note that this assumption is necessary since for example it is false if $n = 8$), then $x^2 = 1 \pmod{p}$ implies $x = 1$ or $-1 \pmod{p}$. Ie, 1 and -1 are the only elements mod p that equal their own inverse.

Proof. $x^2 - 1 = 0 \pmod{p}$ implies $(x+1)(x-1) = 0 \pmod{p}$. This means that in the product $(x+1)(x-1)$, p must appear in the prime factorization of either $x+1$ or $x-1$, since we showed earlier that if p divides a product ab then p divides a or p divides b . Therefore x is either -1 or 1 mod p as desired.

□

Theorem. (Wilson's theorem) Let p be an odd prime, then $(p-1)! = -1 \pmod{p}$.

Proof. Suppose p is an odd prime. $1 * 2 * 3 * \dots * (p-1)$ is a product of units, and the terms come in unit-inverse pairs unless they are self-inverse. We want to determine which of these are self-inverse. But by the lemma, we see that the only things that are self inverse are 1 and $p-1$, so everything else goes into unit-inverse pairs which multiply to 1. Therefore they all cancel mod p and we are left with $p-1$.

□

We will now investigate when a number squares to give -1 mod p . This will be the final theorem of section 3.3.

Example: When $p = 5$, 2 squares to 4 which is $-1 \pmod{5}$. When $p = 7$, you cannot square a number and get 6 mod 7 – You can see this by trying all the possibilities. When $p = 13$, 5 squares to 25 which is $-1 \pmod{13}$. No luck when $p = 19$. There is actually a pattern here.

Theorem. If p is an odd prime, then -1 is a square mod p if and only if p is congruent to 1 mod 4.

Proof. Proof: Suppose $p \equiv 1 \pmod{4}$. Then by Wilson's theorem, I know that -1 is congruent to $(p-1)! \pmod{p}$. We can write $(p-1)!$ as $(1)(2)(3) \dots \left(\frac{p-1}{2}\right) \left(\frac{p+1}{2}\right) \dots (p-2)(p-1)$. Lets subtract p from all terms in the second half of the expression. You can see that this will not change the value mod p , and that it becomes $(1)(2)(3) \dots \left(\frac{p-1}{2}\right) \left(-\frac{p-1}{2}\right) \dots (-2)(-1)$. But since $p \equiv 1 \pmod{4}$, there are an even number of these minus signs so we can cancel them. Therefore we have $\left(\left(\frac{p-1}{2}\right)!\right)^2 \pmod{p}$ is -1. So we have constructed such a number.

As an example, for $p = 29$, $x = 14!$ is a number that if you square it gives you something that is -1 mod 29.

Conversely, suppose $p \equiv 3 \pmod{4}$, or equivalently that $p \equiv -1 \pmod{4}$. Suppose there exists z such that $z^2 = -1 \pmod{p}$. Now lets put $p-3$, which is divisible by 4 (lets call it $4k$), in the exponent. Since $z^{4k} = ((z^2)^2)^k = (-1)^{2k} = 1 \pmod{p}$, we

know that $z^{p-1} = z^{4k+2} = -1 \pmod p$, contradicting Fermat's little theorem.

□

Note: Fermat's little theorem says that $2^p = 2 \pmod p$. Conversely, if we find that $2^n \pmod n = 2$, then there is a good chance that n is prime. However, it is not always the case. However, under 1000, the only composite numbers which pass the test are 341, 561 and 645.

We used 2 as the base, however we could use other bases like 3 or 5 to try to weed out more candidates. But it turns out that some numbers like 561 and 1729 are always a false positive for any base coprime to the number, and these are called Carmichael numbers. We won't prove this here since we won't use it, it's just a cool thing.

3.4 Encryption

We can use the theorems from section 3.3 to do something interesting - We will construct a way to send a message securely such that only someone who has a special key can decode it in a reasonable amount of time.

Let us agree to take a message and convert it to a number somehow. Perhaps by using letters base 26. Then we will use the RSA scheme to encrypt the message in such a way that only someone who has a certain key to decrypt it. Here is how to do that:

First, we think of two very large primes. There are ways to test for large primes, but that is beyond this course. Fun fact: A few years ago I found $260370566^{65536} + 1$ is prime using a computer search. As of this lecture, it is the 9358th largest known prime.

Then let n be the product of these primes so $n = pq$. We will pick an exponent e which should be coprime to $\phi(n)$. We will publish the pair (n, e) , but it is very hard to factor n if you don't know the factors. We will split our message into pieces M which are less than n in such a way that each M is coprime to n . Since p and q are large we can be almost certain that any random number will be coprime to n . We can send the message $M^e \pmod n$, which we can compute quickly by repeatedly squaring $M \pmod n$ to get the big powers without manually multiplying, and since we can reduce mod n , the numbers don't get too big.

To decrypt this, we need to work out the decoding exponent d . By the Fermat-Euler theorem, we want that $ed = 1 \pmod{\phi(n)}$. Since e was coprime to $\phi(n)$ we can find this using Euclid's algorithm. Then we can use this number d we can find $M^{e*d} \pmod n = M^{1+k*\phi(n)} \pmod n$, but by the Fermat-Euler theorem this is just $M \pmod n$. Note that in order to decrypt in this way, we need to know $\phi(n)$ which involves knowing how to factor n and if we know it we could easily factor n . Therefore, decrypting it is equally as hard as actually factoring n (since if we can factor n we can recover $\phi(n)$ and if we can recover $\phi(n)$ it is possible to show that we can find the primes). But when n is large it can take too long for a computer to factor in reasonable time with the fastest algorithms we know, even though finding large primes is much easier.

4 The real numbers

4.1 Construction

Recall how we constructed the natural numbers using the Peano axioms. We can obtain the integers from the natural numbers by allowing for subtraction. Formally, we could view $\mathbb{Z}$ as the set of equivalence classes of $\mathbb{N} \times \mathbb{N}$ where $(a, b)R(c, d)$ if $a + d = b + c$. We can think of (a, b) as $a - b$. We can construct 0 for $(1, 1)$ and $-a$ for $(1, 1 + a)$. We can define the rules multiplication and addition by $(a, b) + (c, d) = (a + c, b + d)$ and $(a, b) * (c, d) = (ac + bd, bc + ad)$. You can check that this works if you want but we all know how arithmetic works in the integers. We can view $\mathbb{Q}$ as the set of equivalence classes in $(\mathbb{Z} \setminus 0) \times \mathbb{N}$ where $(a, b)R(c, d)$ if and only if $ad = bc$, and we think of (a, b) as $\frac{a}{b}$. We can define multiplication, addition, subtraction and division on this using formulas similar to above.

Now try to think about how we might try to construct $\mathbb{R}$. You know what properties you expect it to have, but it is tricky to define it formally.

We can also define an ordering on the rationals in the way we expect. (ie, depending on whether $ad < bc$). Next lecture we will construct the reals - This construction and its properties is fundamental for all of real analysis, which we have

done in previous levels to prove stuff and will continue to do in future levels.

Note that for any 2 rational numbers, we can find one between them. We can do this by taking their average, ie $\frac{p+q}{2}$. The term for this is that the rational numbers are densely ordered. However, this is not enough due to the existence of irrational numbers, for example we know from previous levels that $\sqrt{2}$ cannot be a rational number.

Note: if $\sqrt{x}$ is rational (a/b in lowest terms) and x is an integer then it must be a square number because then we get $x = \frac{a^2}{b^2}$ so $a^2 = xb^2$ so x must be square by comparing prime exponents. Similarly, if $\sqrt[n]{x}$ is rational and x is an integer, x must be a perfect power of n , and in fact $\sqrt[n]{x}$ must also be an integer.

The idea is that $\mathbb{Q}$ has gaps which we need to fill in. Let A be the set of positive rationals p such that $p^2 < 2$. Then we see that A contains no largest element and no least upper bound/supremum (ie, any such supremum would not be in $\mathbb{Q}$ and we need to transcend to the higher realm). The reason is that for any rational number, if we suppose it is the least upper bound, we can find one closer to $\sqrt{2}$, which there are many ways to do, giving us a contradiction. The idea is that we can construct the reals by saying “I have decided that I want the least upper bound of the set A to exist.”, and then the least upper bound to any bounded set of the reals will exist by construction. This is why I have said it is an axiom in the past. This is also how you fill in the gaps in $\mathbb{Q}$.

So here is how we characterize the reals: We start with 0 and 1 and have an operation $+$ and $*$ and an ordering $<$ satisfying the following conditions:

1. $+$ is commutative and associative and adding 0 does nothing
2. $*$ is commutative and associative and multiplying by 1 does nothing and $0 \neq 1$
3. It is always true that $a * (b + c) = a * b + a * c$
4. For every real number, $-x$ exists and if x is not 0, $\frac{1}{x}$ also exists.
5. Adding or multiplying 2 real numbers gives another real number
6. For any a and b , we have exactly one of $a = b$, $a < b$ or $a > b$, and this ordering is transitive and antisymmetric (in the sense that $a < b$ and $b < a$ are mutually exclusive).
7. For any c , $a < b$ implies $a + c < b + c$ and, if $c > 0$, $a < b$ implies $ac < bc$.

Now this gives us all the rational numbers – We can make integers by adding and make rationals by taking reciprocals and adding them together. In fact, the ordering above is unique. For example, suppose $1 < 0$, then $0 < -1$ since we can add -1 to both sides, which means that -1 is positive, so we can multiply both sides of $1 < 0$ by -1 to get $-1 < 0$ which is a contradiction. So we know that $1 > 0$.

Now suppose $1/3 < 0$. Then $2/3 < 1/3$ and $1 < 2/3$ by additivity. This contradicts that $1 > 0$. Also, again by additivity, we know that $0 < 1 < 2 < 3 < \dots$. The point is for any 2 rational numbers we can argue from these axioms in the same way that the ordering is as we expect.

Now we get to the most important property:

8. For any non-empty subset of the reals that is bounded above (i.e. there exists x such that x is greater than or equal to everything in the set), its least upper bound is an element of the reals (i.e. any other upper bound is greater than or equal to x). This is important because we can always define, say, the set of rationals less than π , or the square root of 2, or whatever, then just take the least upper bound, so this is why we have what we intuitively expect the real numbers to be.

Note that non-empty is important or the least upper bound would be negative infinity. And bounded above is important since otherwise the least upper bound would be positive infinity.

Now given the real numbers and the ordering, we can add the square root of -1 and its multiples and its sum with the real numbers to get the complex numbers (See figure 5). A decimal expansion is, for example, we can define $3.14159265\dots$ as the least upper bound of $\{3, 3.1, 3.14, 3.141, 3.1415, \dots\}$.

	Counting numbers, haha $\mathbb{N} = \{1, 2, 3, \dots\}$
	Just two natural numbers! $\mathbb{Z} = \{a - b \mid a, b \in \mathbb{N}\}$
	Just two integers! $\mathbb{Q} = \left\{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\right\}$
	The limits of infinite convergent Cauchy sequences of rational numbers $\mathbb{R} = \left\{\lim_{n \rightarrow \infty} a(n) \mid a : \mathbb{N} \rightarrow \mathbb{Q}, \forall \epsilon > 0, \exists N \in \mathbb{N}, \forall n > N, a(n) - a(N) < \epsilon\right\}$
	Just two real numbers! $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$

Figure 5 (Don't worry about the slightly different jargon as the idea is the same)

We will see in section 5 a sense in which the jump to the real numbers is more substantial than the rest of the jumps.

Also, to construct the reals, we can use Dedekind cuts: they are exactly the least upper bounds of subsets $S \subset \mathbb{Q}$ that are non-empty and satisfy $a \in S, b < a \implies b \in S$, i.e. all rational numbers to the left of a “point”³ on the number line.

In complex numbers we define addition and multiplication as follows

$$- (a, b) + (c, d) = (a + c, b + d)$$

$$- (a, b) * (c, d) = (ac - bd, ad + bc)$$

We can see that the positive integers (as defined by the Peano axioms) and integers constructed from them and rational numbers are in the real numbers by properties 1-7. 1 generates a copy of the integers. By repeatedly adding 1 we do not just get back to where we started because otherwise we would not be able to have the unique ordering as defined above. This is the essence of why we indeed have the integers and the rational numbers contained in the reals.

Example. The least upper bound of both $(0, 1)$ and $[0, 1]$ is 1. In both cases, 1 is an upper bound (everything in the sets is less than or equal to 1) and any smaller number is not an upper bound (since there is a number above it in the set – eg 0.99999 is not an upper bound because of 0.999995 and 1.00001 is not a least upper bound because of 1.000005). Note that the least upper bound is sometimes an element of the set itself and sometimes not, and sets like $(0, 1)$ do not even have a largest element, perhaps contrary to intuition. However, for non empty bounded sets the supremum always exists: for a set S we can write $\sup(S)$ to mean the least upper bound of S , since we had to go all the way through all this and beyond just to prove that A level stats works. (a, b) is called the open interval a to b and $[a, b]$ is called the closed interval a to b . In future analysis courses we will see that open and closed intervals have surprisingly different properties – we have seen this a bit already in previous work.

The set $\{0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots\}$ also has a least upper bound of 1: Numbers of the form $1 - \frac{1}{n}$ get arbitrarily close to 1 but never reach it. We will do some things and then come back to this example.

4.2 The Archimedean property

For this whole section, we will be **very** pedantic.

Proposition. The natural numbers are not bounded above. This is obvious, but we will prove it from the definitions above.

Proof. Suppose on the contrary that the natural numbers are bounded above. Let $c = \sup(\mathbb{N})$. By definition, c is a least upper bound, so $c - 1$ is not an upper bound for $\mathbb{N}$. This means there is a natural number greater than $c - 1$. But then the successor of that number is greater than c , which is a contradiction. So done.

□

³Yes I know saying point is technically circular reasoning, but I'm trying to make a mental picture

We see that the construction of the real numbers is kind of genius in the sense that we need all of these obvious things to actually hold, and to not sneak in a possibility like the natural numbers looping in on themselves (eg if we leave out some of the Peano axioms from earlier it might look ok but then something like the integers mod 7 might satisfy them).

Corollary. For any real number $t > 0$, there exists a natural number n such that $\frac{1}{n} < t$. This and the proposition above is known as the Archimedean property.

Proof. $\frac{1}{t}$ is real since we have defined it to be the case that every non-zero number has a real reciprocal. This is not an upper bound for the natural numbers, so pick a natural number n greater than it. Then consider $\frac{1}{n}$ which must be less than t .

If we're being pedantic, $\frac{1}{n} < t$ since $t^{-1} < n$ and we can multiply both sides by tn^{-1} , which is positive because we can multiply it by n which is positive and then get a positive number t .

□

We have of course seen what it means for a sequence to converge, since we have had to do this to justify some A level maths things, but we will come back to this.

Remark. A set S is bounded below if $-S$ is bounded above. This is obvious. And if this happens S must have a greatest lower bound (minus the least upper bound of $-S$). We can denote a greatest lower bound of S by $\inf(S)$ (I might have shown this before I'm not sure).

Corollary. $\inf\left(\left\{\frac{1}{n} : n \in \mathbb{N}\right\}\right) = 0$ because by the corollary above because 0 is a lower bound and if $t > 0$ is a lower bound this is not possible since we can find something smaller than it. Therefore the set above which we said we would come back to indeed has 1 as its supremum.

Remark. We think the real numbers go on forever, but in fact there are no infinitely large or infinitely small numbers that are real. This is also counterintuitive but it is very important to get used to these counterintuitive ideas.

Note: $\sqrt{2}$ actually exists. The set of rational numbers which squares to less than 2 has a least upper bound. What is this least upper bound squared? We need to prove that it is actually 2. Of course, this number c exists and is between 1 and 2.

Define $A = \{x \in \mathbb{R} : x \geq 0 \text{ and } x^2 < 2\}$ and let $c = \sup(A)$. If $c^2 < 2$ then there is small $t > 0$ with $(c+t)^2 < 2$, contradicting the fact that c is an upper bound.

Note: The technical justification is that

$$(c+t)^2 = c^2 + t(2c+t) < c^2 + 5t < 2$$

This is because $c < 2$, $t < 1$, and we pick t less than a fifth of the difference between c^2 and 2.

A similar argument establishes that if $c^2 > 2$ then for some small $t > 0$, $(c-t)^2 > 2$ contradicting the fact that c is the *least* upper bound.

Note: A similar proof shows that many other gaps in the real numbers are filled.

Note that one can see that if a, b are rational then $a+b$ and ab and $a-b$ and (if b is not 0) a/b are all rational. Also, if ab or $a+b$ is rational for all rational numbers b then a is rational.

Proposition. The rational numbers are dense in the reals. This means they are everywhere, i.e. between any 2 numbers we can find a rational number. Again, this should be obvious to you, but we will prove it.

Proof. Let a to b be an interval which we can shift by an integer until a and b are positive. Then by the Archimedean property there exists an n with $\frac{1}{n} < b-a$. Now let T be the set of natural numbers k such that $\frac{k}{n} \geq a$. This is non-empty because there exists a k with $k \geq bn$ as the natural numbers are unbounded. T is a non-empty set of natural numbers so by the well ordering principle it has a minimal element m . Now consider $\frac{m-1}{n}$. Then we know that $\frac{m-1}{n} < a$ since $m-1$ is

less than the least k such that $\frac{k}{n} \geq b$ by definition. So we just need to show that $\frac{m-1}{n} > a$. Suppose that $\frac{m-1}{n} \leq a$, then $\frac{m}{n} = \frac{m-1}{n} + \frac{1}{n} \leq a + \frac{1}{n} < b$ since we picked n such that $\frac{1}{n} < b - a$, but m is the least element such that $\frac{m}{n} \geq b$, so this is a contradiction as we showed $\frac{m}{n} < b$.

□

Proposition. The irrationals are also dense in this sense.

Proof. Between any two numbers we can find a rational number, and we can find a second rational number between that one and one of our endpoints. Let these rational numbers be p and q . Then $p + \frac{1}{\sqrt{2}}(q - p)$ is between p and q so certainly between our original two numbers, but it is irrational.

□

4.3 Sequences of real numbers

Now we will talk about sequences, which are enumerated collections of real numbers, or formally we can consider them to be functions from the natural numbers to the real numbers. Sequences are often written as $a_1, a_2, a_3, \dots$, or we can write the whole sequence as (a_n)

Recall that a sequence tends to a limit if it eventually gets arbitrarily close to some value and stays that close. The definition is elegant but it took mathematicians a long time to come up with a definition that works.

For example (as we saw in previous levels), the sequence $0, 0.9, 0.99, 0.999, \dots$ does indeed get closer to 35, but we don't want to say 35 is the limit because it does not get arbitrarily close to 35, we clearly want the limit to be 1.

And we do not want things to just get arbitrarily close, we want them to stay that close. If we have a sequence like $1, 2, 1, 1, 2, 1, 1, 1, 2, 1, 1, 1, 2, \dots$ does not stay arbitrarily close to 1. So we want to make it precise that the sequence gets to and stays arbitrarily close to a limit.

A sequence converges to a limit L if for any tolerance $\varepsilon > 0$, no matter how small, we have that there is an N such that for all $n > N$, $L - \varepsilon < a_n < L + \varepsilon$. Then we say that the sequence tends to the limit L . Or we can write that $|a_n - L| < \varepsilon$.

We want to say that if we put a small buffer zone around the limit, the terms in the sequence will eventually go to that buffer zone. As an example, Figure 6 shows what that looks like, we eventually stay in any of the tolerance bands:

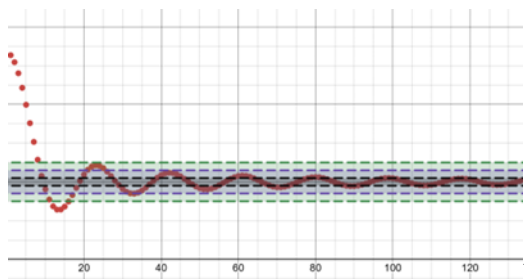


Figure 6

And recall that we want to think of $|a - b|$ as the distance between a and b on our number line or on the complex plane. And recall the triangle inequality: $|a - c| \leq |a - b| + |b - c|$ which says that the distance from A to C is no more than the distance from A to B plus the distance from B to C . So an important trick we will use a lot is $|a - c| = |a - b + b - c| \leq |a - b| + |b - c|$.

A sequence that does not converge diverges. A sequence goes to infinity if $\frac{1}{a_n} \rightarrow 0$, and a sequence goes to positive infinity if for every $M > 0$ all terms are eventually larger than M . A sequence is bounded if there exists an $M > 0$ such that $|a_n| \leq M$ always, but bounded does not imply convergent. However convergent does imply bounded: If L is the limit we fix any ε (say 1) then after a_k the sequence is at most $L + \varepsilon$ so the sequence is bounded after a_k so the whole thing is bounded by $\max(|a_1|, |a_2|, \dots, |a_k|, |L| + 1)$, and similarly bounded below just like it is bounded above.

Example. $0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots \rightarrow 1$

Proof. The difference between a term and 1 is $\frac{1}{n}$, but by the Archimedean property for any ε we can pick an n such that $\frac{1}{n} < \varepsilon$.

□

Example. $0, \frac{1}{2}, 0, \frac{1}{4}, 0, \frac{1}{6}, 0, \dots \rightarrow 0$

Proof. Given $\varepsilon > 0$, pick $n > \frac{1}{\varepsilon}$ by the Archimedean property, then for any $k > n$, $|a_k - 0|$ is either 0 or $\frac{1}{k}$, so less than ε .

□

Example. $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots \rightarrow 0$

Proof. Given $\varepsilon > 0$, pick $n > \frac{1}{\varepsilon}$ by the Archimedean property. $|a_n| = \frac{1}{2^n} < \frac{1}{n} < \varepsilon$ again by the archimedean property.

Proof of the inequality $2^n > n$ implicitly used above: For natural numbers 2^n is the size of the power set of $\{1, 2, \dots, n\}$ which contains more than n elements since it contains $\{1\}, \{2\}, \dots, \{n\}$ plus some more.

□

Example. $(-1)^n = -1, 1, -1, 1, \dots$ does not converge to any limit. This is because for a band of width less than 2, we cannot possibly make the sequence stay there. For all $\varepsilon = 0.5$ for example, this is never going to work: We can never have it be such that both -1 and 1 are both within 0.5 of something.

We can write the definition of convergence as

$$\forall \varepsilon > 0 \exists N \in \mathbb{N}, L \in \mathbb{R} : k > N \implies |a_k - L| < \varepsilon$$

Its negation says there exists an ε such that this is not true, therefore we can write

$$\forall L \in \mathbb{R} \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists k > N : |a_k - L| \geq \varepsilon$$

to mean a sequence does not converge.

Notice how this definition of a limit is more rigorous and precise than the hand wavy “infinitesimal” that past mathematicians like Newton used to use.

Proposition. Limits are unique.

Proof. This is straight forward and intuitive enough that in the past we have used it. But we will prove this from the definition. Assume for a contradiction that a and b are both limits of a sequence and they are different, then for large enough n , $|a_n - a| < \varepsilon$, $|a_n - b| < \varepsilon$.

Idea: We cannot stay simultaneously within two disjoint bands.

We want to set $\varepsilon < \frac{|b-a|}{2}$, then we have $|a_n - a| + |a_n - b| \geq |b - a| > 2\varepsilon$ by the triangle inequality so we cannot have both $|a_n - a| < \varepsilon$, $|a_n - b| < \varepsilon$.

□

Definition. A sequence is eventually bounded if there exists $N, M > 0$ such that for all $n > N$, $|a_n| < M$.

Proposition. Every eventually bounded sequence is bounded.

Proof. It is bounded by $\text{Max}(|a_1|, |a_2|, |a_3|, \dots, |a_N|, M)$.

□

Every convergent sequence is bounded as we proven in my notes from last lecture.

Any unbounded sequence does not necessarily tend to infinity in the limit: A counterexample is a sequence like 0, 1, 0, 2, 0, 3, 0, 4, 0, 5, .

Definition. A sequence is monotonic if it is (not necessarily strictly) decreasing or increasing.

Theorem. (monotone convergence theorem for sequences – the monotone convergence theorem for integrals from level 6 is different) Every bounded monotonic sequence converges.

Proof. We basically did this in level 4. The least upper bound of the sequence exists and must be equal to its limit.

For pedants, if the least upper bound is more than the limit the sequence never gets within some ε of the least upper bound (contradicting it being the least upper bound) and if it is less than the limit the sequence eventually exceeds it (contradicting it being an upper bound)

□

Remark. Boundedness is necessary: 1, 2, 3, 4, ... does not converge. We need a bound so the sequence has something to converge to. We even know that convergence implies boundedness in general.

We know now that the least upper bound property implies the monotone convergence theorem.

Here is another “obvious” theorem

Proposition. If $a_n \leq d$ for all n and $a_n \rightarrow c$ then $c \leq d$. (It is not necessarily the case that if $a_n < d$ and $a_n \rightarrow c$ then $c < d$: consider $-1/n$ with $d = 0$)

Idea: If $c > d$ we will have to get into a band above d which is not possible.

Proof. Let $\varepsilon < |d - c|$. Then $|a_n - c| \leq |d - c|$ implies $a_n > d$ so we have a problem.

□

Proposition. The limit of the sum of convergent sequences is the sum of the limit

Proof. If $a_n \rightarrow a, b_n \rightarrow b$ then eventually $|a_n - a|, |b_n - b| < \frac{\varepsilon}{2}$, so by the triangle inequality $|a_n + b_n - a - b| < \varepsilon$.

□

We could have put ε instead of $\frac{\varepsilon}{2}$ first and got 2ε at the end, but that is ok because it does not change the spirit of the convergence: it preserves the “arbitrarily small” idea.

4.4 Series of real numbers

An infinite sum is defined as the limit of the partial sums. Ie,

$$\sum_{n=1}^{\infty} f(n) = \lim_{r \rightarrow \infty} \sum_{n=1}^r f(n)$$

Provided the limit above actually exists.

We review the geometric series as an example and reproduce the proof that it converges if the common ratio is less than 1 in absolute value and diverges otherwise. The idea was that the partial sum is $\frac{1-r^{k+1}}{1-r}$ which approaches $\frac{1}{1-r}$ if $|r| < 1$ and blows up if $|r| > 1$ and oscillates if $r = -1$ and goes 1, 2, 3, 4, 5, ... if $r = 1$.

Now here is a fun one.

Proposition. $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$

Proof.

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \dots > 1 + \frac{1}{2} + 2\left(\frac{1}{4}\right) + 4\left(\frac{1}{8}\right) + 8\left(\frac{1}{16}\right) + \dots = 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots$$

Which clearly tends to infinity.

□

We can get a good estimate for the terms.

Proposition. Let x be an integer. Then

$$\ln(x) < \sum_{n=1}^x \frac{1}{n} < 1 + \ln(x)$$

Proof. If you're being pedantic you might say we need to define the logarithm. But the truth is that from all the previous levels and your previous knowledge you should know well what the logarithm is. As in level 6 when we proved part of this statement, it is just an area argument. We will use $x = 5$ as an example to show this, and we will use the fact that $\ln(x) = \int_1^x \frac{1}{t} dt$. See figure 7:

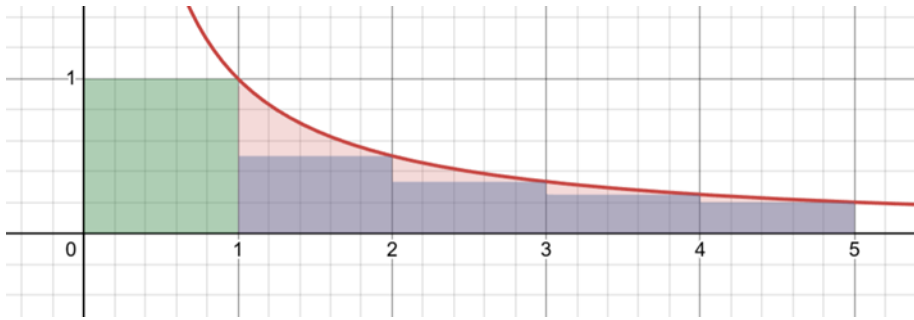


Figure 7

We see that in this image, Red area < Green area (Stack the reds on top of each other), and therefore Red area + Purple area < Green area + Purple area. This inequality is saying exactly that $\ln(x) < \sum_{n=1}^x \frac{1}{n}$. The other side comes from saying that Green area + Purple area < Green area + Red area + Purple area, where it is easy to see that the left hand side is $\sum_{n=1}^x \frac{1}{n}$ and the right hand side is $1 + \ln(x)$ since the green area is 1.

□

What is even more interesting is that the quantity $(\sum_{n=1}^x \frac{1}{n}) - \ln(x)$ which is between 0 and 1 actually converges to a limit as $x \rightarrow \infty$. Its limit is approximately 0.5772. In fact, it is not hard to see why this limit exists: it is, roughly speaking, the total red area in the above image as you extend it to infinity.

Personal story: When I was probably around 5 or 6 years old and playing around with my calculator and tried the $\sum_{n=1}^x \frac{1}{n}$ series, which is known as the harmonic series, I noticed that the sums used in the proof above (which my dad showed me) like $\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}$ or $\frac{1}{9} + \frac{1}{10} + \dots + \frac{1}{16}$ that were known to be at least a half converged to a value around 0.693 and I didn't know what this value was and had never seen it. It turns out from the logarithmic behavior from the proof above that this value is exactly $\ln(2)$.

Lets consider $\sum_{n=1}^{\infty} \frac{1}{n^2}$. We can bound this as follows:

$$1 + \left(\frac{1}{2^2} + \frac{1}{3^2}\right) + \left(\frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2}\right) + \dots < 1 + \frac{2}{2^2} + \frac{4}{4^2} + \frac{8}{8^2} + \dots = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots = 2$$

Where now we are grouping from 2^k to $2^{k+1} - 1$. Therefore since the series is increasing and bounded above by 2 so it converges.

As for what its limit is exactly, this is a surprisingly deep question and the answer is $\frac{\pi^2}{6}$. We do not need to prove this since it's just a cool fact and not something we will build work on – the point of the example was to prove convergence. A proof can be found in the misc results section of the website (this infinite sum is called the basel problem), or when we come to this example in the complex analysis undergraduate sublevel.

Technically to do this more properly we should really get bounds on the partial sums and not manipulate infinite sums.

4.5 Decimal expansions

Let (d_n) be a sequence with $d_n \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Then look at the series $\sum_{n=1}^{\infty} \frac{d_n}{10^n}$. This is the decimal expansion of the number we can write out as $0.d_1d_2d_3\dots$. Now note that the series is bounded below by 0 and bounded above by 1 since the partial sums are increasing and each bounded above by 1. Also the series is bounded by its largest possible value which is the series $\sum_{n=1}^{\infty} \frac{9}{10^n}$, which by a geometric series we can show that this is 1. We showed a proof that this is 1 all the way back in Level 1.

Theorem. Every real number x between 0 and 1 has a decimal expansion. Furthermore, this is (usually) unique.

Proof. As an example, we will try to reach $x = \frac{1}{3}$. We find that $d_1 = 3$ because the number is between $\frac{3}{10}$ and $\frac{4}{10}$. We can carry on by bounding it between consecutive integers divided by 100, then 1000, etc, until eventually we get that it is 0.3333... But the difference between our x and our lower fraction $\frac{3}{10}, \frac{33}{100}, \frac{333}{1000}, \dots$ clearly tends to 0 as it is less than $\frac{1}{10^n}$ so the limit of these partial sums exists and can be found for any real number.

To prove almost uniqueness, suppose that $0.a_1a_2a_3\dots = 0.b_1b_2b_3\dots$. Suppose that k is the first place where they differ and suppose that $a_k < b_k$ since the same idea holds if it is the other way around. Then we know that

$$\sum_{j=k+1}^{\infty} \frac{a_j}{10^j} \leq \sum_{j=k+1}^{\infty} \frac{9}{10^j} = \frac{1}{10^k}$$

Therefore the only possibility is that $b_k = a_k + 1$ since if they differed by more than 1 this tail sum would not be able to differ enough to compensate. In fact, if $b_k = a_k + 1$ then the tails have to differ by $\frac{1}{10^k}$ and thus must be 00000... and 99999.... Therefore decimal expansions are unique unless they end with trailing 0's or trailing 9's or equivalently if they can be written as $\frac{a}{10^b}$ with a and b integers.

□

For example, $0.999\dots = 1$, even though reddit user [u/SouthPark_Piano](#) would disagree.

4.6 Irrational and transcendental numbers

We will now give a different proof of a theorem we proved in level 2.

Theorem. A decimal expansion is periodic (eventually repeats) if and only if the number is rational.

Proof. If it is periodic then we know we can find a fraction as follows (assuming that it starts with 0 since adding an integer doesn't affect it):

$$0.a_1a_2\dots a_nb_1b_2\dots b_kb_1b_2\dots b_k\dots = \frac{a_1a_2\dots a_n}{10^n} + \frac{1}{10^n}(0.b_1b_2\dots b_kb_1b_2\dots b_k\dots) = \frac{a_1a_2\dots a_n}{10^n} + \frac{1}{10^n} \frac{b_1b_2\dots b_k}{10^k - 1}$$

which is clearly rational.

Otherwise, write $x = \frac{u}{v}$. Rewrite this as $x = \frac{p}{2^a 5^b q}$ by factoring the denominator, and do this such that we factor it fully so that q is coprime to 10. Now we have $10^{a+b}x = \frac{p2^b 5^a}{q}$ where a, b, p, q are non-negative integers. Since multiplying by 10 does not affect rationality or periodicity we just need to check that this has a periodic decimal expansion. We will take a remainder so we have $\frac{r}{q}$, and q is coprime to 10 so that means $10^{\phi(q)} - 1 = kq$ for some k by the Fermat-Euler theorem. But then note that $\frac{r}{q} = \frac{kr}{kq} = \frac{kr}{10^{\phi(q)} - 1}$ which is just 0. followed by the digits of kr repeated every $\phi(q)$ places.

□

Now we get to a fun theorem.

Theorem. e is irrational

Proof. Suppose $e = \frac{a}{b}$, then $e = \frac{a(b-1)!}{b!}$ where the numerator is an integer. But then note that $e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{b!} + R$ where R is some error term. Since $e - R$ and e are both an integer divided by $b!$, so is R , but R is $\frac{1}{(b+1)!} + \frac{1}{(b+2)!} + \frac{1}{(b+3)!} + \dots$

We can get a contradiction because we can show that $0 < R < \frac{1}{b!}$ so R cannot be an integer multiple of $\frac{1}{b!}$ as claimed. Since $b \geq 1$,

$$R < \frac{1}{b!} \left(\frac{1}{b+1} + \frac{1}{(b+1)^2} + \frac{1}{(b+1)^3} + \dots \right) \leq \frac{1}{b!} \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots \right) = \frac{1}{b!}$$

So that's our contradiction. □

Theorem. The number $L = \sum_{n=1}^{\infty} \frac{1}{10^{n!}} = 0.1100010000000000000000001000\dots$ is transcendental

Proof. Proof: We will use the fact that for any polynomial p there exists a constant k such that $|p(x) - p(y)| \leq k|x - y|$ for x and y between 0 and 1, and this is because p has a maximum derivative so k is at most that (by the mean value theorem, see level 6 or analysis but anyway its obvious since the derivative is continuous in this case and it measures the slope and you know how derivatives work). We will use also the fact that a polynomial of degree d that is non-zero has at most d real roots (it can't have more than d factors by the factor theorem).

Suppose L is the root of a polynomial p with integer coefficients and suppose this has finite degree d . We see that $0 < L < 1$. We know that there exists a k such that $|p(x) - p(y)| \leq k|x - y|$ for x and y between 0 and 1.

Set $L_n := \sum_{k=1}^n \frac{1}{10^{k!}}$ so that $L_n \rightarrow L$. Note that $|L - L_n| = \sum_{k=n+1}^{\infty} \frac{1}{10^{k!}} \leq \frac{2}{10^{(n+1)!}}$. Now write the polynomial out as $a_d x^d + a_{d-1} x^{d-1} + \dots + a_1 x + a_0$ with $a_d \neq 0$. Notice that $L_n = \frac{s}{10^{n!}}$ for some natural number s , so $p(L_n) = \frac{t}{10^{d \cdot n!}}$ for some integer t . This will be non-zero for sufficiently large values of n since otherwise there would be more than d roots if t is 0 infinitely many times. Therefore, since we are assuming $p(L) = 0$ we have

$$|p(L_n) - p(L)| = |p(L_n)| = \frac{|t|}{10^{d \cdot n!}} \geq \frac{1}{10^{d \cdot n!}}$$

Therefore,

$$\frac{1}{10^{d \cdot n!}} \leq |p(L_n) - p(L)| \leq k|L_n - L| \leq \frac{2k}{10^{(n+1)!}}$$

This implies that for sufficiently large n ,

$$\frac{1}{10^d} \leq \frac{2k}{10^{n+1}}$$

This inequality does not hold for sufficiently large n so we have a contradiction. □

There is a result beyond the course that says $e = \sum \frac{1}{n!}$ is also transcendental.

5 Countability

Definition. A set is said to be countably infinite if there is a bijection between this set and $\mathbb{N}$, and countable if it is countably infinite or finite. This intuitively means we can list or enumerate the elements of the set.

Proposition. $\mathbb{Z}$ is countable

Proof. $0, -1, 1, -2, 2, -3, 3, \dots$

□

Lemma. Any subset of $\mathbb{N}$ is countable.

Proof. Let S be a non empty subset of $\mathbb{N}$. By the well ordering principle there is a least element s_1 . There is also (if there are more elements) a second least element s_2 , then s_3 , etc. If at some point this process ends we have a finite set which is countable. If not, the map $\mathbb{N} \rightarrow \mathbb{N}$ sending $n \rightarrow s_n$ is the bijection we need. This is injective because these minimal elements are all different, and this is surjective because if k is in S then k is a natural number and there are less than k elements in S less than k so we must hit it at some point.

□

Theorem. The following are (obviously) true:

1. X is countable if and only if there exists an injection from $X \rightarrow \mathbb{N}$
2. If X is non-empty, then X is countable If and only if there is a surjection from $\mathbb{N} \rightarrow X$

Proof. 1. If X is finite it obviously injects into $\mathbb{N}$. If X is countable it bijects to $\mathbb{N}$ so it injects into $\mathbb{N}$. Conversely, suppose X injects into $\mathbb{N}$, then trivially X has a bijection to its image s where s is a subset of $\mathbb{N}$ and subsets of $\mathbb{N}$ are countable.

2. If X is countable then either it is finite in which there is a trivial surjection $\mathbb{N} \rightarrow X$ or it is countably infinite so by definition there is a bijection. So suppose f is a surjection $\mathbb{N} \rightarrow X$. So for each a in X pick the least element in $\mathbb{N}$ that maps to a (which exists as it is a surjection and by the well ordering principle) then the union of all such least elements is a subset of $\mathbb{N}$ and is in bijection with X , so X is countable.

□

Note that any subset of a countable set is countable, since the countable set has an injection to $\mathbb{N}$, and said injection with its domain restricted to any subset is still an injection to $\mathbb{N}$.

We can view countability as saying a set is not fundamentally bigger than $\mathbb{N}$, which perhaps will mean the next result is surprising.

Theorem. $\mathbb{N} \times \mathbb{N}$ is countable.

Proof. See figure 8.



Figure 8

The image shows how we can get a bijection if we take the elements connected by the arrows and stretch them out like a rope. Alternatively, we could define a function from $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ that is an injection, as we know that is enough. We can do this by sending (a, b) to $2^a 3^b$. This is an injection by uniqueness of prime factorization.

□

Proof. We did this in level 6. This proof is very famous and to keep these notes self-contained, here is the idea:

Suppose we have a list of real numbers between 0 and 1, then write

$0.a_1a_2a_3\dots$

$0.b_1b_2b_3\dots$

$0.c_1c_2c_3\dots$

Then we get a contradiction because a number which differs by a_1 in the first digit, b_2 in the second digit, c_3 in the third digit, and so on (in a way that avoids non-uniqueness issues discussed earlier), is not in this list.

□

Corollary. We get that transcendental numbers exist and that there are uncountably many of them, by the simple fact that there are only countably many non-transcendental numbers but uncountably many reals.

Recall that we can use a diagonal argument to show that $\mathbb{R}$ (and in fact any interval of real numbers) is uncountable. We can use the same argument to show that a set does not have a bijection to its power set. We will do this proof now.

Theorem. A set can never have a bijection to its power set, the power set is always larger.

Proof. We will do this for $\mathbb{N}$ but exactly the same argument applies to any set.

Suppose, for example, that $\mathbb{N}$ has a bijection to its power set. Then in each subset in our list, each natural number will either be in (Y for yes) or not (N for no) in the list. So as an example we can write it in a table like this:

Y	N	Y	...
N	N	Y	...
N	Y	N	...
...	...	...	...

Now I have put the diagonal elements in bold. Then flip them, so in this example we get NYN... . Then this is not in our list. So contradiction.

□

We can write this formally as $S = \{n \in \mathbb{N} : n \notin S_n\}$ where S_n is the element of our “bijection” and we can apply this to any set. This gives a pretty elegant way to see the inequality $2^n > n$.

In terms of where $|P(\mathbb{N})|$, $|\mathbb{R}|$ and $|\mathbb{C}|$ fall on the “uncountability” hierarchy, it turns out they are all the same size, as we will see.

Note that $|\mathbb{R}| = |(0, 1)|$ because we can do $\tan\left(\pi\left(x - \frac{1}{2}\right)\right)$ as a bijection.

Theorem. $|\mathbb{R}| = |\mathbb{C}|$

Proof. We essentially want to show that $|\mathbb{R}| = |\mathbb{R} \times \mathbb{R}|$. We will use the previous fact to reduce this to showing that $|(0, 1)| = |(0, 1) \times (0, 1)|$.

We will invoke a theorem, which we will prove shortly, that says that if there is an injection $a \rightarrow b$ and an injection $b \rightarrow a$ then there is a bijection $a \rightarrow b$.

To do this, first note that it is clear that $(0, 1)$ injects into $(0, 1) \times (0, 1)$ by sending x to $\left(\frac{1}{2}, x\right)$.

The slightly harder part is to inject $(0, 1) \times (0, 1)$ into $(0, 1)$. We will do some trickery to avoid issues with non-unique expansions.

We will first write some element of $(0, 1) \times (0, 1)$ in binary (base 2) so we have $(0.a_1a_2a_3\dots, 0.b_1b_2b_3\dots)$ by picking always the non-terminating expansion when we have non-uniqueness then we will send this to the real number $0.a_1b_1a_2b_2a_3b_3\dots$ but this will be as a number in base 10, this ensures that a different expansion gives a genuinely different number, since we never have any 9's. This gives us the injection we need.

□

Example. $|\mathbb{R}| = |\mathbb{R} \setminus \{0\}|$ where the bijection is just sending non-negative integers x to $x+1$ and leaving everything else unchanged.

Here is an alternative proof that $P(\mathbb{N})$ is uncountable. We will do this by finding an injection from $(0, 1)$ to $P(\mathbb{N})$ which shows that $P(\mathbb{N})$ must be at least as big.

To do this, write something in $(0, 1)$ in binary or base 2 (Meaning place values differ by a factor of 2 instead of 10 and the possible digits are 0 or 1). Consider the terminating binary expansion if we need to make a choice. If at the k 'th digit there is a 1, make it so k is in the subset of $\mathbb{N}$ that gets mapped to, and otherwise it does not get mapped to. We can think of this as turning 0's and 1's into N's and Y's. As an example, 0.11101 in binary will map to $\{1, 2, 3, 5\}$. This map, which we will call f , is clearly an injection since we cannot have two different numbers have the same binary expansion. Therefore, if $P(\mathbb{N})$ were countable there would be an injection $g : P(\mathbb{N}) \rightarrow \mathbb{N}$ and the composition $g \circ f : (0, 1) \rightarrow \mathbb{N}$ is an injection which contradicts $(0, 1)$ being uncountable.

Note that there is no set of all sets or else it would contain every element in its power set and we know this is not possible.

Example. Consider a family of open pairwise disjoint intervals A_i of the reals. Then this is countable as each interval contains a rational number but then if the set was uncountable we would have uncountably many rational numbers.

Summary of what we've done so far:

X is countable if

1. There is an injection X to $\mathbb{N}$
2. You can enumerate it
3. It is a countable union of countable sets
4. There is a surjection $\mathbb{N}$ to X

X is uncountable if

1. We can run a diagonal argument to show that there is no surjection $\mathbb{N}$ to X
2. X has a surjection to another uncountable set

Intuitively we think of existence of a bijection to mean sets have the same size. It's like we have an equivalence relation of sizes by existence of a bijection.

Intuitively we think of " A injects to B " to mean " A is at most as big as B " and " A surjects to B " to mean " A is at least as big as B ". For this to make sense, we need to formally prove that if A has both an injection and a surjection to B then there is a bijection A to B , and also that that " A injects to B " is equivalent to " B surjects to A ".

Lemma. Given non-empty sets A and B , the existence of an injection f from A to B is equivalent to the existence of a surjection g from B to A .

Proof. Define $g : B \rightarrow A$ by sending $f(a_0) \rightarrow a_0$ for each a_0 in A and for an element not in the image of f sending it to

anything. This gives a surjection g from B to A . Conversely, if g is a surjection, then partition B into the pre-images of each element in A . Then for each a in A pick something in the corresponding pre-image to send it to to get an injection.

□

Theorem. (Cantor-Schroeder-Bernstein Theorem) If there is both an injection (f) and a surjection (t) from $A \rightarrow B$ then there is also a bijection (h). We can interpret this to mean that $|A| \leq |B| \leq |A| \implies |A| = |B|$

Proof. Note that by the previous lemma there is an injection g from $B \rightarrow A$.

Start by letting g^{-1} be a left inverse of g and f^{-1} be a left inverse of f .

For each a in A , pick the element $g^{-1}(a)$ (this is a left inverse of g) in B in the pre-image of A if it exists, then pick the pre-image of that in A if it exists, i.e. $f^{-1}(g^{-1}(a))$, then consider the sequence of these pre-images: $\dots g^{-1}(f^{-1}(g^{-1}(\dots$ and we will call this the ancestor sequence of a , as it is talking about where a came from.

These sequences will either continue infinitely, or have an even number of steps, or have an odd number of steps. Let A_0 be the set of a in A whose ancestor sequences terminate after an even number of steps, i.e. the last point is in our set A . Now let A_1 be the set of a in A that end in an odd number of steps, and A_∞ be those such that the ancestor sequence does not terminate. Do the same for B . Now we want to show that we can construct bijections between the following partitions: $A_0 \rightarrow B_1, A_1 \rightarrow B_0, A_\infty \rightarrow B_\infty$, then we can construct a bijection out of those three.

f must biject $A_0 \rightarrow B_1$: it is an injection because f is an injection, and it is a surjection because everything in B_1 has an odd number of ancestors so for each b in $B_1, f^{-1}(b)$ exists and is in A_0 , so we have a surjection as b was arbitrary.

Now note that for every a in A_1 , we must have that $g^{-1}(a)$ exists and is in B_0 , and $g^{-1} : A_1 \rightarrow B_0$ is a surjection because each b comes from $g(b)$ and an injection because if $a \neq b$ but they are both in A_1 which is in the image of g (by definition of A_1) then since g is a function we must have that $g^{-1}(a) \neq g^{-1}(b)$.

Now we have a bijection $A_\infty \rightarrow B_\infty$ by f , which is clearly an injection, and it is a surjection because for each b in B_∞ by definition there is an $f^{-1}(b)$ in A_∞ . This was a lot, but we can write our bijection as follows:

$$h(x) = \begin{cases} f(x) & x \in A_0 \\ g^{-1}(x) & x \in A_1 \\ f(x) & x \in A_\infty \end{cases}$$

□

Example. We have a bijection $[0, 1] \rightarrow [0, 1] \cup [2, 3]$ because we have an injection by the identity map. and an injection $[0, 1] \cup [2, 3] \rightarrow [0, 1]$ by sending every x to $\frac{x}{3}$.

Since I have previously mentioned the possibility of “mapping the real numbers in a one to one fashion to the sets of two real numbers” in level 5 when making fun of the system, I figure I should show how to actually do it. It is from the proof above that reals have a bijection to the complex numbers.

And that is the end of the course.